



Sonar: Production Risk Intelligence

Five independent detectors, three chains, 127 vaults — continuously running

Alvaro Alonso

Senior Executive Engineer, Sail

April 2026

Abstract

Sonar is Sail's real-time DeFi risk detection system. It runs continuously and autonomously, in parallel with the Optimal Engine. Its single responsibility is to populate a shared security document with two lists before every rebalancing decision: vault addresses the engine must skip, and stablecoin tokens the engine must not touch. No human is in the loop.

The system deploys five independent detection layers. The Collateral Risk Monitor reads lending market utilization, available liquidity, and borrow rate pressure directly on-chain via batched RPC calls across all six lending protocols and three chains, and computes per-user and system-wide exit feasibility. The Whale Exit Monitor listens to ERC-4626 withdrawal events in real time and flags large exits that presage liquidity stress. The Deep Depeg Scanner replaces lagged price aggregators with Chainlink oracle reads, Uniswap v3 TWAP cross-checks, and DEX pool reserve imbalances, anchored by recursive token dependency trees that resolve every vault to its underlying primitive asset exposure. The GitHub Anomaly Detector polls the six core protocol repositories for emergency commits, abnormally fast pull request merges, and off-hours releases. A Risk Aggregator unifies all detector outputs through a three-tier cooldown system before writing the two engine-facing kill lists. The vault universe currently spans 127 enabled vaults across Base, Arbitrum, and Ethereum mainnet.

Contents

1	Overview	3
2	The Engine Contract	3
3	Vault Universe	4
4	Architecture	5
4.1	Continuous pipeline	5
4.2	Detector ownership	5
5	Detection Layers	6
5.1	Collateral Risk Monitor	6
5.2	Whale Exit Monitor	7
5.3	Deep Depeg Scanner	8
5.3.1	Token Dependency Trees	8
5.3.2	DEX Pool Imbalance	8
5.3.3	TWAP Oracle Sanity Check	9
5.3.4	Composite Depeg Score	9
5.4	GitHub Anomaly Detector	10
6	The Risk Aggregator and Cooldown System	10
6.1	Three-tier cooldown	11
6.2	Vault risk scores	11
7	Signal Coverage Summary	11
8	Speed Architecture	12
9	Ethereum Mainnet Expansion	13
10	Risks and Mitigations	14
11	Next Steps	14
12	Conclusion	15

1 Overview

Sail’s Optimal Engine rebalances stablecoin positions across 127 vaults on three chains continuously, chasing the best risk-adjusted yield. The profitability of that engine depends entirely on whether the pools it selects are safe at the moment of rebalancing. Sonar is the system that answers that question.

Sonar runs as a parallel pipeline, continuously, with no human intervention required between runs. It reads on-chain state directly — the same state that liquidation bots and informed traders read — and translates it into a shared security document before the engine’s next decision cycle. When a lending market approaches full utilization, when a whale exits a vault, when a DEX pool absorbs a disproportionate flow of a weakening stablecoin, or when a protocol pushes an emergency hotfix to its main branch, Sonar detects it and updates the engine’s kill list ahead of the next decision cycle.

Table 1: Sonar at a Glance

Dimension	Production State
Cadence	Continuous, always on
Detection latency	<1 block (~1s on-chain, seconds for GitHub)
Chains	Base, Arbitrum, Ethereum mainnet
Vaults monitored	127 enabled
Depeg signal	Chainlink + Uniswap TWAP + DEX pool imbalance
Depeg resolution	0–90 composite score per token per chain
Collateral visibility	Recursive tree to primitive asset, all 127 vaults
Lending market health	Utilization, borrow rate, available liquidity
Exit feasibility	Per-user and aggregate across all Sail positions
Whale exits	Real-time ERC-4626 Withdraw event watching
Protocol emergency	GitHub monitoring, 4 signal types
Cooldown logic	Three-tier: 1h / 24h / manual sign-off
Engine contract changes	Zero

Sonar reads the same on-chain state that liquidation bots read, at the same latency. It translates that state into vault disables before any aggregator, any social feed, and before most bots act.

2 The Engine Contract

The Optimal Engine applies two security gates before any vault is considered for rebalancing. These gates read a single security document updated by Sonar on every run:

1. **Vault-level kill switch.** If a vault address appears in the disabled-vault registry, the vault is skipped entirely on every subsequent rebalancing cycle.
2. **Token-level kill switch.** The disabled-token registry maps chains to token symbols. Every vault on a flagged chain using a flagged token is excluded in one operation.

Every Sonar detector ultimately writes into one or both of these two registries. The engine reads nothing else from the security document. This constraint is the reason that five detectors across three chains can be added without any engine code changes.

Table 2: Risk Signal to Engine Effect

Risk event	Registry updated	Engine effect
Lending market utilization critical	Disabled-vault registry	Skip vault address
Exit feasibility critically low	Disabled-vault registry	Skip vault address
Borrow APR spike	Disabled-vault registry	Skip vault address
Large single withdrawal	Disabled-vault registry	Skip vault address
Protocol GitHub emergency	Disabled-vault registry	Skip all protocol vaults
Token depeg (Chainlink confirmed)	Disabled-token registry	Block token on chain
DEX pool TWAP divergence critical	Disabled-token registry	Block token on chain
Depeg composite score critical	Disabled-token registry	Block token on chain

3 Vault Universe

The current monitored universe spans 127 enabled vaults across three chains. The most significant recent addition is Ethereum mainnet, which adds 88 vaults dominated by the Morpho MetaMorpho V1 ecosystem — curated vaults from Steakhouse, Gauntlet, Relend, ClearStar, Hyperithm, Kiln, Karak, and others. Morpho’s TVL on Ethereum mainnet currently exceeds \$3 billion across all curators.

Table 3: Enabled Vault Universe by Protocol and Chain

Protocol	Chain	Vaults	Tokens	Type
Aave v3	Arbitrum	2	USDC, USDT	Lending market
Aave v3	Base	2	USDC, EURC	Lending market
Aave v3	Ethereum	2	USDC, USDT	Lending market
Compound v3	Arbitrum	2	USDC, USDT	Lending market
Compound v3	Base	1	USDC	Lending market
Compound v3	Ethereum	2	USDC, USDT	Lending market
Euler	Arbitrum	8	USDC, USDT	Lending + Earn vaults
Euler	Base	1	USDC	Earn vault
Fluid	Arbitrum	2	USDC, USDT	fToken lending
Fluid	Base	2	USDC, EURC	fToken lending
Moonwell	Base	3	USDC, EURC	Lending market
Morpho	Arbitrum	9	USDC, USDT	MetaMorpho vaults
Morpho	Base	13	USDC, EURC	MetaMorpho vaults
Morpho	Ethereum	84	USDC, USDT, EURC	MetaMorpho V1 vaults
Total		127	USDC, USDT, EURC	

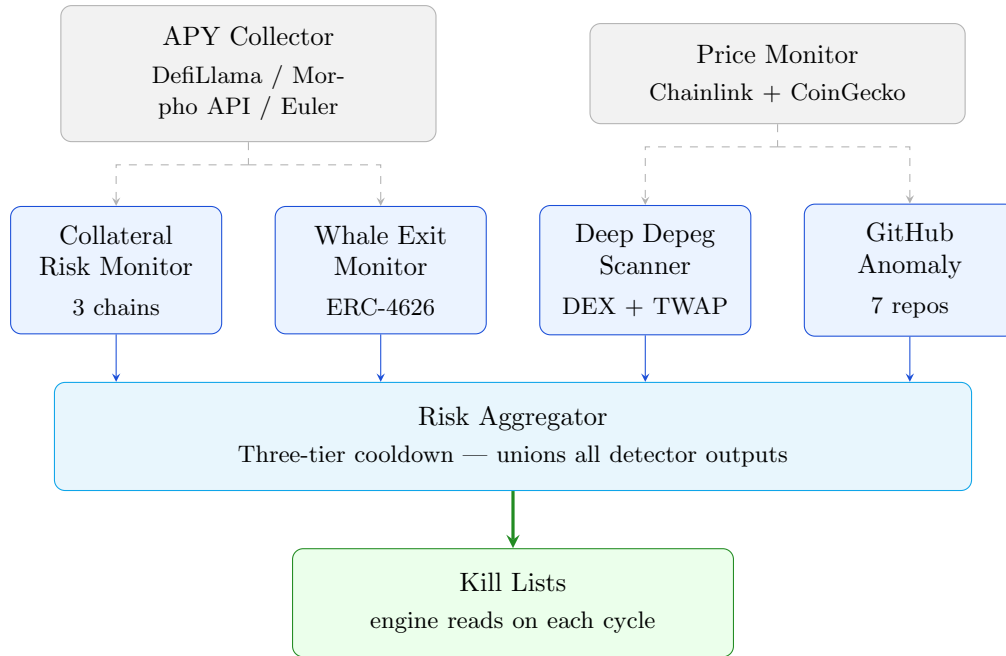
Two stablecoin markets, one pipeline

Sail operates two independent stablecoin markets: USD (USDC, USDT) and EUR (EURC). These markets are never mixed in the engine or in Sonar’s monitoring logic. DEX imbalance pairs are same-currency only — USDC/USDT on Arbitrum and Ethereum, never USDC/EURC. EUR monitoring expands to a DEX pair once a same-currency EUR counterpart becomes available on Base or Ethereum.

4 Architecture

4.1 Continuous pipeline

Sonar runs as a six-step pipeline in a continuous loop. Steps 1 and 2 handle yield data collection. Steps 3 through 6 are the risk detectors. Step 6 always runs last: it reads all detector outputs and rebuilds the engine’s two kill lists with cooldown logic applied.



4.2 Detector ownership

Each of the four detection layers owns a private internal registry that tracks its current output independently. The Risk Aggregator reads all four private registries on every run and rebuilds the engine-facing kill lists. No detector writes to the kill lists directly. This prevents a detector clearing its own output from accidentally removing entries that a different detector legitimately placed.

Table 4: Detector Ownership and Tier

Detector	Kill list updated	Cooldown tier
Depeg Security	Disabled-token registry	Tier 3 (manual)
Collateral Risk	Disabled-vault registry	Tier 2 (24h)
Whale Exit Monitor	Disabled-vault registry	Tier 1 (1h)
Deep Depeg Scanner	Disabled-token registry	Tier 2 (24h)
GitHub Anomaly	Disabled-vault registry	Tier 3 (manual)

5 Detection Layers

5.1 Collateral Risk Monitor

What it does. On every run, reads the health of every enabled lending market via a batched on-chain call: supply utilization, available liquidity, accumulated borrows, and borrow rate. The same call retrieves Sail’s aggregate position sizes to answer the sharper question: *can Sail’s position exit right now?*

Why it matters. High utilization means withdrawals may be throttled or blocked. When a market approaches 100% utilization and the engine needs to exit, there is no available liquidity to receive the position. The exit feasibility check makes this concrete — it is not a utilization ratio but a direct comparison of available liquidity against Sail’s actual exposure.

Table 5: Protocol-Specific On-Chain Read Method

Protocol	Primary contract calls	Key signals
Aave v3	<code>aToken.totalSupply()</code> , <code>asset.balanceOf(aToken)</code> , reserve data provider (opt.)	utilization, available cash, borrow APR
Compound v3	<code>cToken.totalSupply()</code> , <code>cToken.totalBorrow()</code>	utilization, reserves
Euler	<code>eVault.totalAssets()</code> , <code>eVault.totalBorrows()</code>	utilization
Fluid	<code>fToken.totalAssets()</code> , <code>balanceOf(liquidityLayer)</code>	utilization, available cash
Moonwell	<code>getCash()</code> , <code>totalBorrows()</code> , <code>totalReserves()</code> , borrow rate	liquidity, borrow APR

All reads execute as a single `aggregate3` call on the `Multicall3` contract with `allowFailure=True`: a revert on one vault does not break the batch.

Signal thresholds are configurable per deployment. Each metric triggers a two-stage response:

- **Utilization** — a high-utilization WARN stage allows monitoring without action; a critical-utilization stage disables the vault. Thresholds are set conservatively relative to the kink point of each protocol’s interest rate model.
- **Available liquidity floor** — an absolute minimum is enforced independently of utilization,

scaled to aggregate exposure. A vault that cannot absorb a full exit is disabled regardless of its utilization ratio.

- **Borrow APR spike** — a sudden rate jump signals that the market has reached its kink and that capital is leaving. Critical threshold is calibrated per protocol.
- **Exit feasibility ratio** (available liquidity / total exposure) — a **WARN** ratio triggers per-user alerting; a critical ratio disables the vault and writes an exit-urgency record for every affected user.

Systemic exit check. On each run, the total Sail exposure across all user positions is aggregated per vault:

$$\text{systemic_feasibility} = \frac{\text{available liquidity}}{\sum_{\text{all users}} \text{position size}}$$

This catches the concentration risk case: Sail accumulates large exposure relative to a small pool, even when no single user position triggers the per-user threshold.

Protocol notes. Fluid’s available cash lives at a central Liquidity Layer contract shared across all fToken deployments — reading the fToken itself yields zero. Aave v3.1/v3.2 on Base reverts on the reserve data provider; a version-agnostic fallback path using raw token balances works across all Aave versions. EURC has 6 underlying decimals despite the vault share token using 18-decimal precision; a hardcoded decimal map is used for all liquidity math.

5.2 Whale Exit Monitor

What it does. On every run, fetches all ERC-4626 **Withdraw** events for every enabled vault since the last recorded block, per chain. A large single-transaction withdrawal is a leading indicator of liquidity stress: informed actors exit positions before a problem becomes public, often 15 to 60 minutes before any price aggregator reflects it.

Mechanism. The ERC-4626 event signature is:

```
Withdraw(sender, receiver, owner, assets, shares)
```

The **assets** field is decoded directly from the log data and divided by the token’s underlying decimal precision to yield the amount withdrawn. This is compared against the vault’s current TVL from the yield data cache loaded at startup.

The monitor processes up to 999 blocks per **eth_getLogs** call, resuming from the last processed block checkpoint. Cold starts backfill up to 3 hours of history automatically. On Ethereum, where blocks arrive approximately every 12 seconds, each pipeline cycle covers roughly 25 blocks; on Arbitrum and Base blocks arrive much faster.

Signal thresholds are expressed as a fraction of vault TVL and are configurable per deployment. A moderate withdrawal fraction triggers a **WARN** and begins monitoring; a large withdrawal fraction triggers **CRITICAL** and adds the vault to the disabled-vault registry immediately.

EUR token decimal handling

Circle’s EURC (deployed on Base) has 6 underlying decimals, consistent with USDC. The Morpho steakEURCv vault on Ethereum holds a distinct EUR token called EURCv whose withdrawal event `assets` field uses 18-decimal precision. Treating EURCv as 6-decimal produces a 10^{12} magnitude error — a \$12 withdrawal becomes a \$12 trillion alert. The monitor maintains a hard-coded decimal map for all known token variants; EURCv is explicitly mapped to 18 decimals.

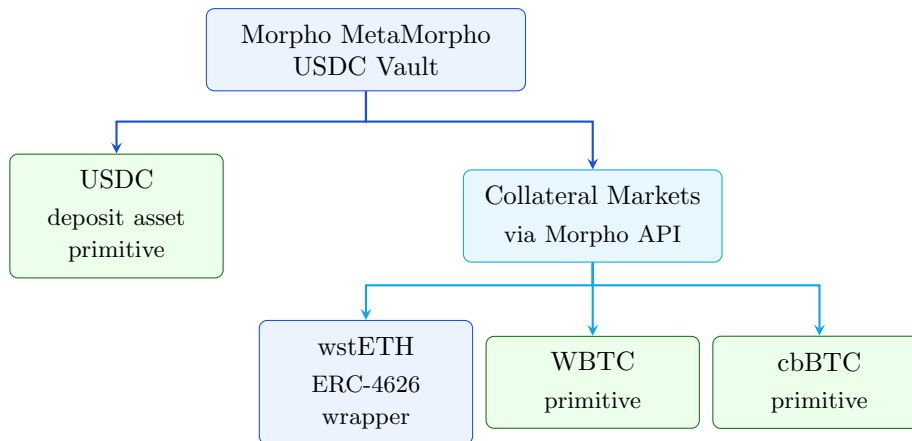
5.3 Deep Depeg Scanner

This layer performs two independent functions: building and caching recursive token dependency trees (on a 30-minute throttle), and running DEX pool imbalance and TWAP oracle sanity checks on every run.

5.3.1 Token Dependency Trees

For every vault in the monitored universe, the tree builder resolves risk exposure to primitive assets — tokens with direct Chainlink price feeds that are not wrappers of anything else. The primitive set includes USD stablecoins (USDC, USDT, DAI, USDE, USDS), EUR stablecoins (EURC, EURCv), ETH and its liquid staking derivatives (wstETH, cbETH, rETH, weETH and others), and BTC wrappers (WBTC, cbBTC, tBTC, LBTC).

For Morpho MetaMorpho vaults, the Morpho API is queried for market allocations: each market exposes its collateral asset (the token borrowers post as collateral). For each collateral token, the builder reads the ERC-4626 `asset()` method to unwrap one level, then recurses until a primitive is reached. Non-ERC-4626 tokens with known underlyings (staked USDS, staked USDe, syrupUSDC, Pendle PTs) are resolved via a hardcoded override table.



127 trees are currently stored. On each run the cached trees are used by the scoring layer: if a collateral token depegs, every vault with exposure to that collateral propagates the signal upward.

5.3.2 DEX Pool Imbalance

Stablecoin liquidity pools on Uniswap v3 are designed to hold paired tokens in roughly equal proportions. When one stablecoin begins losing its peg, informed traders swap out of the weakening

token. The pool absorbs those swaps and ends up holding a disproportionate share of the depegging asset. This imbalance appears on-chain often 15 to 60 minutes before any price aggregator reflects the move.

Pool addresses are discovered at startup via the Uniswap v3 Factory `getPool()` across three fee tiers (100, 500, 3000 bps). Token balances are read via `IERC20.balanceOf(pool)`, batched in a single Multicall3 call.

Pairs monitored (same currency only):

- Arbitrum: USDC / USDT0
- Ethereum: USDC / USDT, USDC / DAI, USDC / USDE
- Base: none (EURC has no same-currency counterpart yet)

Pool imbalance is measured as the dominant token’s share of total reserves. A moderate imbalance triggers a **WARN** contribution to the composite score; a severe imbalance triggers a **CRITICAL** contribution. Pool imbalance alone cannot disable a token — it requires corroboration from a price or TWAP signal.

5.3.3 TWAP Oracle Sanity Check

The Chainlink spot price (read in the same Multicall3 batch) is compared against the Uniswap v3 30-minute TWAP for every monitored pair. A significant divergence between the two indicates either oracle manipulation (a Chainlink update delayed after a sharp on-chain move) or an unabsorbed price shock. Neither condition is visible to aggregators.

The TWAP is derived from the `observe([1800, 0])` call:

$$\overline{\text{tick}} = \frac{\text{tick}_t - \text{tick}_{t-1800}}{1800} \quad \text{TWAP price} = 1.0001^{\overline{\text{tick}}}$$

adjusted for the decimal difference between the two tokens. All reads are in the same Multicall3 batch as the balance reads — zero additional RPC calls.

A small divergence between Chainlink spot and the 30-minute TWAP triggers a **WARN**. A larger divergence triggers a **CRITICAL** fast-path disable: both tokens in the pair are added to the disabled-token registry immediately, without waiting for the composite score to accumulate.

5.3.4 Composite Depeg Score

All depeg signals for each token are combined into a per-chain composite score:

$$\text{score} = \underbrace{\text{price deviation}}_{0-40 \text{ pts}} + \underbrace{\text{pool imbalance}}_{0-25 \text{ pts}} + \underbrace{\text{TWAP delta}}_{0-15 \text{ pts}} + \underbrace{\text{cross-chain flag}}_{0-10 \text{ pts}}$$

Cross-chain: if the same token is flagged on two or more chains simultaneously, it receives a bonus on every chain — a simultaneous multi-chain stress event is systemic, not a bridge arbitrage artefact. A moderate total score triggers **WARN**. A high score combined with a corroborating price deviation or TWAP signal triggers **CRITICAL**: the token is added to the disabled-token registry.

Guardrail: pool imbalance alone, without a corroborating price or TWAP signal, cannot reach a **CRITICAL** disable.

5.4 GitHub Anomaly Detector

Protocol engineering teams respond to exploits on GitHub before making any public statement. Emergency pull requests are opened and merged in under an hour. Hotfix commits land on `main` with keywords like “pause”, “exploit”, or “vulnerability”. Release tags appear at 03:00 UTC. All of these are detectable in seconds via the GitHub Events API — well within any pipeline cycle.

Repositories monitored:

- aave/aave-v3-core, compound-finance/comet, moonwell-fi/moonwell-contracts-v2
- morpho-org/morpho-blue, morpho-org/metamorpho
- euler-xyz/euler-vault-kit, Instadapp/fluid

Four signal types, all from a single Events API call per repository:

1. **Emergency keyword commit.** A push to `main` whose commit message contains any of: emergency, pause, exploit, hotfix, vulnerability, incident, freeze, hack, attack, drain, security, reentrancy, overflow.
2. **Direct push to main.** A commit pushed without going through a pull request — code that bypassed the normal review cycle.
3. **Fast pull request merge.** A PR opened and merged in under 2 hours — abnormal velocity consistent with an emergency response.
4. **Off-hours release.** A release tag published between 00:00 and 06:00 UTC — the pattern of a hasty or automated deployment.

Rate limiting. The detector polls only when more than 32 minutes have elapsed since the last check. At approximately 1.8 polls per hour across 7 repositories, the system uses roughly 13 requests per hour — well within the 5,000 req/hr budget with an authenticated token.

Output. On any signal, all vault addresses for the affected protocol are added to the disabled-vault registry under a Tier 3 (manual sign-off) cooldown.

Why GitHub is faster than Twitter for emergency detection

A protocol team pushes an emergency fix to GitHub before posting anything publicly. The window between the first hotfix commit and the first public tweet is typically 10 to 60 minutes, based on historical incidents across Euler, Compound, and Aave. GitHub monitoring closes that window.

- Events API reflects a push within seconds.
- False positive rate is low: emergency keywords on `main` are rare in normal operation; off-hours releases are intrinsically suspicious.

6 The Risk Aggregator and Cooldown System

The Risk Aggregator is the final step in every pipeline run. It reads all four detector private registries, applies tier-based cooldown logic, and writes the two engine-facing kill lists. No detector writes to the kill lists directly.

6.1 Three-tier cooldown

Not all risk signals carry the same information value or decay at the same rate. A whale withdrawal resolves quickly as liquidity providers fill the gap. A GitHub emergency signal may indicate a compromised protocol and should require human confirmation before the vault is re-enabled.

Table 6: Cooldown Tiers

Tier	Auto-expiry	Detectors	Rationale
1	1 hour	Whale Exit Monitor	Self-correcting signal. Liquidity returns within hours as LPs respond.
2	24 hours	Collateral Risk, Deep Depeg	Market stress dissipates within a trading day. Borrow rates revert; DEX imbalances arbitrage out.
3	Manual	Depeg Security, GitHub Anomaly	Structural signals. A confirmed depeg or a protocol emergency requires human sign-off.

Cooldown mechanics. For each entry in a detector registry:

- If still flagged this run: refresh the timestamp, keep in engine kill list.
- If no longer flagged: hold for the tier’s cooldown period, then expire automatically (Tier 1 and 2) or wait for manual deletion (Tier 3).
- Re-enable: delete the registry entry. If the underlying condition has resolved, the detector will not re-flag on the next run.

The cooldown state is persisted to survive pipeline restarts.

6.2 Vault risk scores

The aggregator also writes a per-vault risk score record containing the contributing detectors, highest severity level, and timestamp of first detection. This is purely observational — the engine ignores it.

7 Signal Coverage Summary

All thresholds are configurable per deployment and calibrated to aggregate exposure. The table below shows the signal type, severity stages, and engine effect for each detector.

Table 7: Detection Signals and Engine Effect

Detector	Signal	Stages	Kill list	
Depeg Security	Peg deviation from oracle	WARN CRIT	/	Disabled-token
Collateral Risk	Lending utilization	WARN CRIT	/	Disabled-vault
Collateral Risk	Exit feasibility ratio	WARN CRIT	/	Disabled-vault
Collateral Risk	Borrow APR spike	CRIT		Disabled-vault
Collateral Risk	Available liquidity floor	CRIT		Disabled-vault
Whale Monitor	Single withdrawal / TVL	WARN CRIT	/	Disabled-vault
Deep Depeg	DEX pool reserve imbalance	WARN CRIT	/	Score only
Deep Depeg	TWAP vs Chain-link divergence	WARN CRIT	/	Disabled-token
Deep Depeg	Composite score	WARN CRIT	/	Disabled-token
GitHub Anomaly	Emergency key-word commit	CRIT		Disabled-vault
GitHub Anomaly	Fast PR merge	CRIT		Disabled-vault
GitHub Anomaly	Off-hours release	CRIT		Disabled-vault
GitHub Anomaly	Direct push to main	CRIT		Disabled-vault

8 Speed Architecture

The pipeline must detect and register an exploit signal before the engine’s next rebalancing decision. Source selection is entirely determined by latency.

Table 8: Data Source Latency and Role

Source	Latency	Role	Cost
Web3 direct (Multicall3 batched)	~1 block (<2s)	All on-chain checks	Existing
Chainlink oracle reads	0.5% deviation trigger	Primary price truth	Free
ERC-4626 event logs	~1 block	Withdrawal detection	Free
Uniswap v3 pool reads	~1 block	Imbalance + TWAP signal	Free
GitHub Events API	Seconds after push	Protocol emergency	Free
CoinGecko API	60s+ aggregate lag	Depeg fallback (EURC)	Free
Morpho GraphQL API	Seconds to minutes	Collateral tree building	Free
DefiLlama / Yearn	Minutes	Yield data only	Free

Multicall3 as the performance foundation. Every on-chain check — lending market state, pool balances, TWAP ticks, Chainlink prices — executes as a single `aggregate3` call on the Multicall3 contract (`0xcA11bde...CA11`, deployed identically on every EVM chain). A single RPC round-trip fetches all state for an entire chain.

Chainlink reads, TWAP checks, and lending market state are all within one block of any on-chain event. Sonar detects and registers a signal before the engine makes its next decision.

9 Ethereum Mainnet Expansion

Ethereum mainnet is the most significant recent addition to the monitored universe, bringing 88 new vaults. The majority are Morpho MetaMorpho V1 curated vaults spanning the full landscape of Ethereum yield curators: Steakhouse Financial, Gauntlet, Relend, Resolv, ClearStar, Hyperithm, Kiln, Karak, and others. Morpho’s MetaMorpho V1 TVL on Ethereum currently exceeds \$3 billion.

Protocol coverage on Ethereum. Collateral risk monitoring covers Aave v3 and Compound v3 using the same batched RPC patterns established for Arbitrum and Base. The 84 Morpho MetaMorpho vaults are covered by the dependency tree builder and DEX monitoring; individual Morpho Blue market collateral utilization is tracked via the Morpho API.

Block time and event coverage. Ethereum’s 12-second block time means approximately 25 blocks are covered per pipeline cycle. The whale monitor processes these in a single `eth.getLogs` call per run. Cold starts backfill up to 3 hours.

EUR stablecoins on Ethereum. The Morpho steakEURCv vault accepts deposits of EURCv, a distinct 18-decimal EUR token. There is currently no Chainlink EURC/EUR feed on Ethereum mainnet and no same-currency EUR DEX pair in the monitored universe; this vault has collateral

risk and withdrawal event monitoring but no active DEX imbalance tracking. EURCv is classified as an independent EUR primitive in all dependency tree resolution.

10 Risks and Mitigations

- **False positives triggering unnecessary exits.** A false critical signal adds a vault to the disabled registry and the engine exits the position at the next cycle, incurring gas and forfeited yield. Mitigation: Tier 1 false positives auto-resolve in 1 hour; Tier 2 in 24 hours; Tier 3 fires only on Chainlink-confirmed depegs and GitHub emergencies, which have inherently low false-positive rates. All signals are persisted to observability records for post-hoc diagnosis and threshold tuning.
- **Morpho API availability.** Dependency trees for MetaMorpho vaults depend on the Morpho GraphQL API. If the API is unavailable, trees are stale but all on-chain checks continue against the cached version. Trees carry a build timestamp; trees older than 24 hours generate a log-level alert without disabling vaults.
- **RPC rate limits.** Every new on-chain check uses Multicall3 batching. A typical full-chain run requires one `eth_getLogs` call and one Multicall3 call per chain, regardless of vault count.
- **Pool imbalance false positives from arbitrage.** Uniswap v3 pools can briefly show imbalance during arbitrage sweeps. The TWAP guardrail prevents these transient spikes from triggering disables: an imbalance signal alone cannot reach the critical composite threshold without a corroborating price deviation or TWAP divergence.
- **Concentration risk.** Aggregate exposure to small-pool vaults may approach available liquidity even when no individual position breaches the per-user exit threshold. The systemic exit check surfaces this proactively: the WARN feasibility ratio provides a buffer before the CRITICAL disable threshold is reached.

11 Next Steps

1. **Threshold calibration from live data.** Current thresholds were set conservatively from first principles. After four weeks of production data on three chains, statistical anomaly detection (mean $\pm$ 2 standard deviations per vault per metric) should layer on top of absolute thresholds to suppress idiosyncratic market microstructure noise.
2. **EURC DEX monitoring on Ethereum.** When a same-currency EUR stablecoin DEX pair becomes available on Ethereum, EURC can be promoted to active DEX imbalance and TWAP monitoring. The infrastructure already handles the chain; only a pool address and a Chainlink EUR/USD feed are needed.
3. **Social sentiment layer (Phase 5).** DeFiLlama hacks feed, DeFiHackLabs exploit database, and a curated four-tier account list (security firms, protocol official accounts, curators, collateral issuers). Social signals trigger an immediate deep scan but cannot directly disable vaults without on-chain corroboration.
4. **Historical baseline collection.** Per-vault normal withdrawal distributions, borrow rate baselines, and pool imbalance means should be collected passively over 30 days. This enables z-score gating to complement the current absolute threshold model.

12 Conclusion

Sonar is a production system that runs continuously, without human intervention. Five independent detection layers cover 127 vaults across three chains: lending market health, withdrawal event watching, DEX pool reserve monitoring, TWAP oracle cross-checking, recursive collateral tree resolution, and GitHub emergency signal detection. Every signal is on-chain or sourced directly from protocol repositories — the same data that liquidation bots and informed traders read, at the same latency.

The engine contract has not changed since Sonar was first deployed: two kill lists, two gates, zero code changes required as the vault universe scales. The three-tier cooldown system makes the pipeline self-correcting for transient market events while remaining conservative on structural signals. Tier 1 and Tier 2 events auto-resolve within hours or a trading day; Tier 3 events stay blocked until a human clears them.

The expansion to Ethereum mainnet — 88 new vaults covering the full Morpho MetaMorpho V1 curator ecosystem — positions Sail as one of the few automated yield systems with production-grade, multi-detector risk coverage across the three dominant EVM yield chains simultaneously.

Every signal — lending market utilization, whale withdrawal, DEX pool imbalance, TWAP divergence, or an emergency commit on GitHub — translates automatically into a vault address on the engine’s skip list.

Five detectors. Three chains. 127 vaults. No human in the loop.